



How do Process Mining Users Act, Think, and Feel?

An Explorative Study of Process Mining Use Patterns

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Abstract Process mining, a family of techniques for analyzing large amounts of data collected on business processes, has gained significant practical and academic importance. Extant process mining research mainly examines technical aspects. Only recently has research started to investigate organizational aspects of process mining, such as how organizations use process mining to create business value. On an individual level, research has examined the analysis strategies of individual process mining analysts. So far, however, the literature does not provide a holistic investigation of individual process mining use, including individuals' behavior, cognition, and affective states. Yet, understanding individual process mining use is pivotal for realizing its organizational value. To address this shortcoming, this paper examines the individual use of process mining employing a multiple case study with process mining users from six large organizations in a post-adoption context, i.e., in organizations that have several years of experience with process mining. Based on the configuration of how process mining users act, think, and feel in practice, four distinct process mining user categories were identified: (1) process mining influencers, (2) power users, (3) process participants, and (4) strategic users. For practitioners, the findings provide insights into the actual process mining use of individuals and what shapes their use patterns. This information

enables tailoring process mining training and algorithms to specific user categories.

Keywords Process mining · Information systems use · Use patterns · User categorization · Post-adoption

1 Introduction

Process mining is a big data analytics technology (Chen et al. 2012) that enables organizations to extract process knowledge from event logs (van der Aalst et al. 2011). It offers new affordances that allow organizations to improve process efficiency, monetary key performance indicators, and non-monetary performance indicators such as employee security (Badakhshan et al. 2022). According to a forecast by Fortune Business Insights, the global process mining revenue is expected to grow from 2.46 billion US dollars in 2024 to 46.39 billion US dollars in 2032 at a compound annual growth rate of 44.3 percent per year (Fortune Business Insights 2024). Furthermore, process mining vendors worldwide report record sales and market capitalizations, e.g., the German company Celonis has a current market capitalization of 13 billion US Dollars (Celonis 2022).

While the majority of research on process mining today is still concerned with developing novel or improving existing process mining techniques (e.g., Augusto et al. 2018), there is an increasing research interest in how process mining is used in organizations. In this respect, scholars have called to investigate process mining on different levels (vom Brocke et al. 2021). On an organizational level, studies have examined how process mining enables transparency over business processes (Eggers et al. 2021), how organizations use process mining to create

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business value (Badakhshan et al. 2022) as well as success factors for process mining projects (Mans et al. 2013; Mamudu et al. 2024). Further, studies have pointed out that process mining can pose specific challenges, such as coping with induced process transparency (Grisold et al. 2020), data quality, governance, and collaboration (Syed et al. 2020).

On an individual level, use patterns “encompassing an individual’s emotions, cognitions, and behaviors while employing an information technology (IT) to accomplish a work-related task” (Ortíz de Guinea and Webster 2013, p. 1166) with process mining have so far not received systematic attention from research. A notable exception in this regard are Zerbato et al. (2021) and Zerbato et al. (2022), who examine the working practices of process analysts. However, they focus on the user group of analysts only and mainly consider cognition and behavior. This is in line with the general literature on the use of information systems (IS) that mainly focuses on behavior but tends to neglect affects and cognition of individuals (Straub and del Giudice 2012). As process mining provides transparency over organizational work (Badakhshan et al. 2022; Eggers et al. 2021) and potentially individuals (Mannhardt et al. 2018; Grisold et al. 2020), which can lead to privacy concerns, it is important to capture process mining use comprehensively. Understanding the different users who employ process mining with respect to behavior, cognition, and affect is important, as the nature of technology use considerably impacts organizational performance and returns from IT investments (Chen et al. 2015; Devaraj and Kohli 2003). From a practical perspective, identifying user categories is important, as it allows the targeted development of process mining algorithms, training, and general user support.

Moreover, most studies investigate the initial organizational adoption of process mining (e.g., Grisold et al. 2020). However, to understand individuals’ actual use of process mining technology, we need to examine their use during the organizational post-adoption phase. This is necessary to ensure that the system is actually used by employees (Jaspersen et al. 2005) and that initial set-up issues, such as poor data quality, do not bias the emotions, cognitions, and behaviors of users engaging with process mining.

This leads us to pose the following research questions: (1) *How are process mining users’ behaviors, cognitions, and affects configured?* (2) *What user categories emerge from these process mining use patterns?*

To address these questions, we adopt a qualitative research approach. This is consistent with prior work on the use of process mining in organizations (Badakhshan et al. 2022; Eggers et al. 2021) and allows us to deep-dive into the contextual circumstances that lead to and influence

process mining use. Specifically, we conducted an explorative multiple case study that included six organizations (Eisenhardt 1989; Yin 2018). All companies are in the post-adoption phase (Cooper and Zmud 1990), i.e., they have considerable experience with process mining, and process mining is widely diffused across the organization. Overall, we conducted and analyzed 31 interviews with a variety of stakeholder groups that have access to process mining technology. Based on our abductive analysis, we derive four distinct user categories: process mining influencers, power users, process participants, and strategic users that differ with respect to behaviors, cognitions, and affects in relation to process mining technology. We point to antecedents, such as training offers, individual background, and organizational culture, that facilitate the emergence of the respective use patterns. Our study adds to the literature on post-adoption and the use of process mining in organizations. For organizations, the derived use patterns are relevant as they allow tailoring process mining training to a specific user category to support optimal usage. Our research is also valuable for technical research on process mining. Our findings can be used for the enhancement of existing and the development of new process mining techniques, as they allow academics and practitioners to better understand the unique use and associated requirements of a specific user category.

The remainder of this article is organized as follows. First, we summarize the literature on process mining and IS use. We then describe how we conducted our multiple case study. Afterward, we describe our findings. In particular, we present the use patterns for each of the four process mining user categories. Next, we discuss our findings and propose future research avenues. We conclude the paper by summarizing its key contributions to the body of knowledge.

2 Conceptual Background

2.1 Process Mining

Process mining is a big data analytics technology that harnesses large data volumes from event logs stored in IS (Chen et al. 2015; Badakhshan et al. 2022). Based on these data, process mining applications support the discovery, monitoring, and improvement of business processes (van der Aalst et al. 2011). As such, it shifts the focus from how processes should be to how they are actually performed (Badakhshan et al. 2022).

While process mining research has traditionally focused on technical aspects, an emerging body of research explores the use of process mining in organizations (Eggers et al. 2023; Badakhshan et al. 2022; Grisold et al. 2020;

Zerbato et al. 2022). Most importantly, process mining creates transparency that helps organizations generate process awareness (Eggers et al. 2021) and, ultimately, business value by taking action on the business process transparency created by process mining (Eggers and Hein 2020; Wurm and Claus 2024). Previously, organizations relied on interview- or workshop-based approaches to obtain insights about their business processes. Such procedures typically suffer from subjective gaps in knowledge and a lack of objective validation (Dumas et al. 2018). In contrast, process mining is based on event logs and, therefore, allows users to derive a more objective and complete view of the process. It can depict the process flow in (near) real-time with ever-continuing updates (Kerremans et al. 2021). Process mining users can zoom into particularly interesting or problematic behavior to identify root causes (Badakhshan et al. 2022). Enabled by process transparency (Eggers et al. 2021), process mining allows organizations to (1) perceive end-to-end process visualizations and performance indicators, (2) make sense of process-related information, (3) make decisions based on data instead of gut feelings, and (4) implement interventions to improve business processes (Badakhshan et al. 2022). Because real-time data connectivity leads to immediate updates of process visualizations, interventions in processes can be quickly evaluated and adapted, if necessary (Badakhshan et al. 2022). Overall, this allows organizations to speed up the business process lifecycle (Dumas et al. 2018) and improve process efficiency, monetary key performance indicators, and non-monetary performance indicators such as employee security and customer satisfaction (Badakhshan et al. 2022).

The literature further suggests how organizations should conduct process mining projects. Previous research proposes methods for how process mining projects can be executed (Aguirre et al. 2017; van Eck et al. 2015) and how processes can be redesigned with process mining (Mărușter and van Beest 2009). Success factors for process mining projects include process mining expertise, project management, and resource availability (Mans et al. 2013).

On an individual level, researchers have conceptualized different categories of users based on anecdotal evidence. These groups include process analysts, process participants, process stakeholders, and system engineers (vom Brocke et al. 2021). Scholars have cautioned that increased transparency induced by process mining might lead to distrust and perceived surveillance on an individual level, which should be addressed by developing appropriate coping mechanisms (Grisold et al. 2020). Furthermore, scholars have investigated the analysis strategies of individual process analysts using process mining when given a specific analysis task (Zerbato et al. 2022, 2021). Generally speaking, users run through the phases of understanding,

planning, analyzing, and evaluating. This line of work provides insights into the rational choices that users make to carry out their work. However, what remains unexplored is the actual use of process mining applications within organizations after initial adoption, especially regarding behavior, cognition, and affect. Understanding process mining use can help organizations leverage the full and long-term potential that this technology offers.

2.2 Information Systems Use

IS use is one of the most influential and mature streams in IS research as it is linked to individual, organizational, and societal impact (Burton-Jones et al. 2017). It is generally understood as “an individual user’s employment of one or more features of a system to perform a task” (Burton-Jones and Straub 2006, p. 231). The relevance of IS use in practice and its trajectory in research is rooted in the fact that it is the nexus where IS investments translate into business value (Devaraj and Kohli 2003). A prerequisite for IS use is organizational IS adoption. Research on organizational IS adoption (e.g., Cooper and Zmud 1990) distinguishes three high-level stages: pre-adoption activities, adoption decision, and post-adoption activities. In order to realize the associated benefits of IS, use has to move toward the post-adoption stage, in which a system is actually used by employees (Jasperson et al. 2005; Bagayogo et al. 2014).

IS use in the pre-adoption phases is often operationalized with “lean” measures (Burton-Jones and Straub 2006). Lean measures refer to simple, basic, or surface-level indicators of user engagement with a system. They often refer to the use amount, such as frequency, duration, or feature deployment. Lean measures do not delve deeply into how effectively or meaningfully the user interacts with a system. Over the years, conceptualizations and measurements of IS use have become more sophisticated and “richer” (Burton-Jones et al. 2017). Where previous research solely relied on behavioral components (Straub and del Giudice 2012), scholars have added cognitive and affective components to move away from descriptive and lean measurements (Subramani 2004). Richer measures capture more nuanced insights into how users engage with a system and aim to understand the quality of interaction. We follow this richer view of IS use, as it more adequately captures the complex nature of IS use in practice.

Behavior describes the users’ “doing” (Burton-Jones and Gallivan 2007), which can be studied along dimensions of quantity and quality (Jain and Kanungo 2005). Quantity-focused indicators can encompass use duration (Venkatesh et al. 2003) or frequency (Davis et al. 1989). However, an increased duration or frequency is not necessarily linked to greater benefits (DeLone and McLean 2003), but the

quality of use is pivotal. Researchers have assessed the depth and breadth of feature employment to evaluate the quality of use behavior, also referred to as deep structure use (Zhang 2017). Other use behaviors that have been studied are extended use (i.e., additional feature and function use to support a task) (Hsieh and Wang 2007; Hsieh et al. 2011) or enhanced use, referring to novel ways of applying IS features (Bagayogo et al. 2014). Actual IS use behavior can also be investigated in relation to the task that is (intended to be) executed. Here, studies have considered the number of tasks that can be accomplished using a system (Igbaria et al. 1997). Use can also comprise resistance behaviors (Lapointe and Rivard 2005) and non-use (Kane and Labianca 2011). Non-use can also be included as behavior, as experience with a certain system in a post-adoption stage can also lead to decreased or diminished interaction. Insights into non-use behaviors can thus present valuable perspectives into the post-adoption activities of an individual.

“**[C]ognition** (i.e., thinking) means the mental process of knowing, including aspects such as perception, reasoning, and judgment” (Kim et al. 2007, p. 511). Researchers argue that IS use can be described as a series of conscious, rational decisions based on expectations and evaluations of consequences (Ortíz de Guinea and Markus 2009; Karahanna et al. 1999). The levels of cognitive intensity can lie on a continuum from no or negligible cognitive effort to a great deal of cognitive effort (Bagayogo et al. 2014). Little cognitive extent may be reflected by automaticity, associated with habit formation. A habit requires minimal mental effort (Limayem et al. 2007). Especially in the post-adoption phase, habits are crucial as they are generally associated with positive outcomes, freeing cognitive resources for other tasks and ultimately increasing individual performance (El-Khatib and Barki 2012; Ortíz de Guinea and Markus 2009).

Affect refers to what a user feels (Burton-Jones and Gallivan 2007). Unlike conscious thinking and evaluations, affect refers to emotions (Agarwal and Malhotra 2005). Often, researchers do not distinguish between affect and cognition, yet “affective responses can occur without any cognitive processing or deliberation” (Zhang 2013, p. 262). Many affect-related phenomena have been recognized to occur with IS in the workplace, such as fear and anxiety (Brown et al. 2004), or enjoyment and pleasure (Kim et al. 2007). To synthesize emotions associated with IS use, Beaudry and Pinsonneault (2010) developed a framework that classifies different types of emotions. It includes achievement emotions (e.g., happiness, pleasure), challenge emotions (e.g., excitement, playfulness, flow), loss emotions (e.g., anger, dissatisfaction, frustration), and deterrence emotions (e.g., anxiety, fear). At the same time, scholars have highlighted the occurrence of mixed

emotions, i.e., different types of emotions occurring simultaneously when users interact with IS (Stein et al. 2015).

Taken together, these three dimensions form a use pattern, that is, “a configurational construct encompassing an individual’s emotions, cognitions, and behaviors while employing an IT to accomplish a work-related task” (Ortíz de Guinea and Webster 2013, p. 1166). Use patterns are configurative, as different values and levels of emotions, cognitions, and behaviors occur together. Moreover, the values of use patterns alter over time as context and user perceptions change (Ortíz de Guinea and Webster 2013). For example, unexpected events such as glitches can change how users think about a system and trigger negative emotions, leading to behavioral changes, impacting task performance, and causing a differently configured IS use pattern (Ortíz de Guinea and Webster 2013). Specific cognition, affect, and behavior configurations can portray and distinguish a particular context or group. Hence, configurations can be used to differentiate user categories. Table 1 presents an overview of the dimensions of IS use patterns, their explanations, as well as examples from the process mining literature.

In the context of process mining, recent studies provide selective insights into different dimensions of individuals’ process mining use. Scholars have analyzed the process mining interaction of process analysts, which is a specific organizational role, and identified their cognitive strategies and behavioral actions (Zerbato et al. 2022). Several types of behavioral inconsistency-in-use, such as a lack of data accuracy or variations in individuals’ output interpretations, were identified in a qualitative case study, which hinder effective process mining deployment (Eden et al. 2021). Moreover, studies quantitatively investigate process mining use as behavioral interaction in relation to algorithm appreciation and aversion, identifying an external origin as a driver for the trust and behavioral reliance of process mining tools in conformance checking tasks (Lohoff and Engert 2024). This task perspective has been previously highlighted as relevant in further understanding process mining use (vom Brocke et al. 2021). Regarding cognition, it has been shown that organizational stakeholders perceive process mining differently. While managers in a manufacturing firm appreciate process mining-induced transparency, shopfloor workers might consider it a threat (Stert et al. 2021). Others have identified user commitment as an antecedent to realizing value from process mining (Eggers et al. 2023). Previous studies on process mining use also include affective notions, as its application could be associated with surveillance fears due to the induced process transparency (Eggers et al. 2021; Grisold et al. 2020).

Table 1 Individual process mining use dimensions (based on Burton-Jones and Gallivan 2007)

IS use pattern dimension	Explanation	Examples in the process mining context
Individual <i>behavior</i>	What a user <i>does</i>	Behavioral reliance on the suggestion of a process mining tool in conformance checking tasks (Lohoff and Engert 2024)
Individual <i>cognition</i>	What a user <i>thinks</i>	Cognitive strategies for the use of process mining (Zerbato et al. 2022)
Individual <i>affect</i>	What a user <i>feels</i>	The transparency process mining creates might be perceived as a threat (Stertzt et al. 2021)

As to the adoption stage, previous research on process mining has targeted early adoption stages, characterized by a launch-and-learn approach without expert guidance, data, and governance support (Syed et al. 2020). Prior studies provide initial acceptance obstacles or enablers in the early stages of user interaction with process mining tools or stay at the organizational application level (Thiede et al. 2018). Regarding real-life interaction, some studies collect data from academics with limited process mining experience in organizational practice (Zerbato et al. 2021) or provide participants with fictional or experimental scenarios (Zerbato et al. 2022; Lohoff and Engert 2024). While these studies are valuable in providing insights into specific behavioral mitigators, e.g., tool origin (Lohoff and Engert 2024) or specific use strategies (Zerbato et al. 2022), they lack knowledge about actual application beyond initial thoughts and experimentation in an organizational context.

In summary, studies often only consider future or potential process mining users, target one specific user group (Zerbato et al. 2022), or investigate the use concept one-dimensionally, e.g., only targeting users' cognition. A holistic and context-specific investigation of users' use patterns in a post-adoption phase can contribute further insights into process mining use in organizations and how to translate process mining investments into value (vom Brocke et al. 2021).

3 Methodology

In this section, we detail the methodological approach we followed to answer the research questions. First, we describe the research design; second, the data collection process; and finally, the data analysis.

3.1 Research Design

To explore individuals' process mining use patterns, we adopted a multiple case study methodology following an exploratory approach (Dubé and Paré 2003). We chose this method for three reasons. First, the use of process mining in organizations is still in its early stages, with the

technology having only recently gained traction but spreading quickly across various industries. Case studies are especially well-suited for investigating such emerging phenomena (Eisenhardt 1989; Yin 2018). Second, IS use patterns are complex and interconnected across different users. Employing the case study approach allowed us to receive in-depth insights from informants complemented by secondary material. Third, an individual's use pattern configuration of behavior, cognition, and affect is deeply influenced by the specific context. The case study approach allowed us to examine process mining in its real-world setting, making it particularly effective for capturing the context-dependent dynamics of process mining use (Yin 2018; Miles and Huberman 1994).

To select the cases (see Table 2 for an overview), we followed a theoretical sampling approach relying on the principles of similarity and contrast (Miles and Huberman 1994). Similarities were emphasized across cases regarding the aspects of (1) post-adoption status, (2) process mining vendor, and (3) size and scope. (1) As this study aims to investigate IS use beyond initial adoption, the organizations must have already implemented and actively employed process mining during our data collection period. Organizations that implemented process mining four or more years ago were selected. Thereby, we wanted to ensure that initial implementation issues were addressed and that the organizations gained considerable experience with process mining. (2) Also, all organizations employ the same process mining solution (pseudonymized as *ProMi*), which allowed us to rule out differences in use solely caused by differences in employed software. However, CongloCo B uses a second process mining software (pseudonymized as *VeTwo*) in addition to *ProMi*. (3) All companies in our sample can be classified as large organizations based on employee size, revenue, and operating scope. This is advantageous as there is more differentiation in large organizations concerning process mining-related roles, which allowed us to identify distinct types of users.

At the same time, the cases selected differ in terms of (1) the industry they operate in, (2) the organizational structures and process mining governance, and (3) the business processes for which process mining is applied. (1) The case

Table 2 Case studies overview

	TeleCo	SkyCo	MediCo	AutoCo	CongloCo A	CongloCo B
<i>Industry</i>	Telecommunications	Aviation	Pharma & Crop	Automotive	Conglomerate	Medical Technology
<i>Company size</i>		2021 revenue: > €15 billion				
<i>Employees</i>		> 60 000 (as of 2022)				
<i>Scope</i>		Multinational; operations in Germany				

organizations differ in regard to the industries in which they operate, as Table 2 displays. (2) Further, the case organizations govern their process mining endeavors in different ways and rely on heterogeneous structures to steer process mining. For instance, some case companies such as CongloCoA have established a process mining Center of Excellence (CoE). Other companies, such as MediCo or CongloCoB, did not have a CoE in place at the time of the interviews. Also, the process mining CoEs, if existent, are structurally located in different departments. The CoE at SkyCo is located in the IT department and the CoE at TeleCo in a service center. (3) Moreover, the organizations selected as cases are different regarding their process mining use cases. For instance, CongloCoA uses process mining especially to analyze their *Order-to-Cash* and *Purchase-to-Pay* processes, SkyCo for the *Ground Handling* process of airplanes, and AutoCo for the *Manufacturing* processes.

For transparency reasons, it is important to mention that at the time of the interviews, the first author was employed at CongloCoA. This permitted the author to become a process mining user herself. Thereby, she became part of CongloCoA's process mining community, which included dedicated training. When she engaged in casual conversations about other users' experiences, thoughts, and feelings about process mining, she took notes, which were used for triangulation and validity checks during the data analysis.

3.2 Data Collection

In the first six months of 2022, we conducted 31 qualitative interviews with 28 informants (three follow-up interviews) via online meeting applications, via telephone, or in person. The informants were acquired via the authors' networks and professional social media channels. The interviews were conducted in German or English, depending on the informants' preferences, and lasted 44 min on average. To select the individual informants, we applied two selection criteria. First, we selected the informants in a way that created a diverse sample to cover a "variety of voices" (Myers and Newman 2007, p. 22). Our sample encompasses a variety of organizational roles,

including department heads, IT experts, process mining CoE or Center of Competence members, process managers and operators, as well as internal and external process mining consultants. This means the sample covers a variety of different use contexts and tasks, presupposing varying individual backgrounds. We started our data collection by interviewing individuals who explicitly carried the term "process mining" in their job title. Then, we used snowballing and asked previous informants for less obvious process mining users, such as operative process participants. The latter were more difficult to identify as they typically do not carry the term "process mining" explicitly in their job title. Our sample is also diverse with respect to the location of informants. While most informants were employed in Germany, data was collected from informants in other European countries, Asia, and the Americas. Moreover, our sample covers male and female informants and all age groups.

As the second selection requirement, all informants had to be at least trained to use process mining to have sufficient knowledge to employ basic process mining features. Further, they needed to have access to a process mining tool. Despite technical access and the ability to employ process mining, informants could have been non-users at the time of the interview. For instance, informant SkyCo5 received training and has a license to access the process mining tool, yet he did not operatively use process mining. That is because the process he currently performs is not recorded digitally. Therefore, *years of process mining experience* is indicated with zero years for this informant (see Table 3). Since the literature also acknowledges that IS use can comprise non-use, we found it insightful to interview nonactive users as well. While they did not use process mining actively, they still had an opinion and emotions about process mining use in their organizations. Also, this gave valuable insights into antecedents that hamper process mining use, such as lack of training, lack of digital maturity, or absence of process mining support in the form of a CoE for process mining.

On average, the informants had four years of experience with process mining and had been with their respective organizations for eleven years. This allowed them to offer

Table 3 Informants overview and secondary data sources

Company	Informant number	For the case study relevant position of informant	Tenure	Years of process mining experience	Interview duration (minutes)
TeleCo	TeleCo1	Lead of Process Mining CoE	11	3	53
	TeleCo2	Lead of Process Mining Center of Competence in Procurement	10	4	44
	TeleCo3	Business Analyst Procurement	8	4	69
	Secondary data: use demonstration during interview (1), case study (1), newspaper articles (3)				
SkyCo	SkyCo1	Head of Business Development	5.5	4	36
	SkyCo2	Expert in Process Digitalization	3	7	35
	SkyCo3	Team Lead Ground Operations	23	4	40
	SkyCo4	Expert in Fleet Management	25	3	23
	SkyCo5	Expert in Flight Operations Support	7	0	17
	Secondary data: newspaper articles (2), presentation (1), blog entries (2)				
MediCo	MediCo1	Engineer Automation Solutions	12	9	31
	MediCo2	Business Analyst Procurement	2	2	56
	MediCo3	Senior Director Procurement	20	1	31
	Secondary data: case study (1)				
AutoCo	AutoCo1	CoE Subject Lead for Production & Logistics	5	5	44
	AutoCo2	Quality Reporting IT	16	6	35
	AutoCo3	Production Expert	3	3	39
	AutoCo4	Project Manager in Quality Management	10	4	54
	AutoCo5	Senior Execution Management Consultant	1.5	1.5	56
	Secondary data: case study (1), presentation (1), newspaper articles (3)				
CongloCoA	CongloCoA1	Head of Central Process Mining IT	21	8	38 + 14
	CongloCoA2	Head of Operational Excellence	19	6	102
	CongloCoA3	Business Excellence Professional	21	5	36
	CongloCoA4	Process Mining Analyst	10	4	39
	CongloCoA5	Operational Team Leader Order Management	3	2.5	57
	CongloCoA6	Senior Process Consultant	34	3	76 + 64
	CongloCoA7	Digitalization Consultant	4	0.5	31
	CongloCoA8	Digitalization Consultant	0.5	3.5	21
	Secondary data: demonstrations during interviews (3), case study (1), newspaper article (1), blog entry (1), internal and external presentations (2), press releases (2), informal exchanges (2)				
CongloCoB	CongloCoB1	Subject Matter Expert Procurement	19	7	59 + 56
	CongloCoB2	Senior IT Manager	15	4	36
	Secondary data: demonstration during interview (1), case studies (2), informal exchange (1)				
ProMi	ProMi1	Customer Value Architect	2	2	32
	ProMi2	Customer Value Architect	0.5	0.5	39

valuable contextual insights into both pre- and post-adoption phases. To further triangulate our data and gather additional background information, we also conducted two interviews with consultants from the process mining vendor ProMi. Table 3 displays an overview of the informants.

Most interviews were conducted with process mining users of CongloCoA, which was necessary to grasp the complexity of process mining undertakings within this organization caused by its conglomerate status, i.e., the

organization operates in diverse industries. Moreover, CongloCoB was carved out from CongloCoA in 2018. Hence, process mining was already implemented when the carveout took place. As CongloCoB is majority-owned by CongloCoA, both organizations are still interrelated, yet they are separate legal entities. Also, CongloCoA and B have different process mining structures. For instance, CongloCoA has a dedicated CoE for process mining with a headcount of 30. In contrast, CongloCoB does not have a

CoE. Therefore, we consider CongloCoA and B distinct cases, although the historical evolution of process mining is interrelated.

We conducted semi-structured interviews with primarily open-ended questions (Myers and Newman 2007) to gain insights into process mining use patterns. The notion of IS use following Ortíz de Guinea and Webster (2013) guided the preparation of our semi-structured interview guideline. Several parties participated in preparing the interview guideline, including the author team and a process mining consultant from an external consultancy. Moreover, we conducted a test interview with a manager for big data analytics products. We iteratively adapted the interview guideline based on the experiences gathered during the interviews. In unraveling how and what process mining use patterns occur in practice, the following areas of inquiry were addressed: (1) individual background, (2) how the organization structurally embeds process mining, (3) individual tasks performed with process mining, (4) behavior, thoughts, and feelings towards process mining. In particular, we questioned the informants about their use patterns in the following way. First, to gather information about informants' behavior, we asked them which tasks they (aim to) perform with process mining, how often and how long they use process mining, and which features they apply (e.g., only dashboard use or also more advanced features such as the process explorer, etc.). Second, to get an impression of what informants think, we asked them to what extent the system could live up to their expectations, for instance in terms of the performance of the software. Also, we asked whether they actively need to think when using process mining or whether their use does not require much cognitive effort. Third, we inquired about positive and negative affective states. Therefore, we asked informants how they personally felt when interacting with process mining and what their attitude is about the software and the transparency it creates. The interviews were recorded and transcribed verbatim (Miles et al. 2014). The interview guideline is available in Appendix A (available online via <http://link.springer.com>).

We triangulated the interview data with direct observations of informants interacting with process mining software (Eisenhardt 1989). Those direct observations helped validate the informants' statements. Informal conversations and observations were recorded as field notes (Dubé and Paré 2003). Other secondary data sources include case studies from the process mining provider and the case companies, websites, or newspaper articles. The secondary data were all publicly accessible or sent to us by the informants. We used them primarily to present facts (e.g., size of the company, date of process mining adoption) and the evolution of process mining in the organizations (e.g., use cases for process mining). This means we

used the secondary data mainly for the within-case descriptions and to cross-check the validity of the informants' statements. We collected all relevant data in a case database (Yin 2018).

3.3 Data Analysis

To discover individual process mining use patterns, we entered the analysis with the a priori specification of IS use following Ortíz de Guinea and Webster (2013). That means we adopted the notion of IS use comprising the dimensions of behavior, cognition, and affect and combined this with the insights emerging from our data. During data analysis, we continuously moved back and forth between inductive and deductive reasoning, i.e., we followed an abductive approach (Suddaby 2006; Dubois and Gadde 2002; Timmermans and Tavory 2012). Abduction involves iteratively switching between the data and theoretical notions, representing a non-linear approach to case study research. This method allows researchers to systematically integrate insights from both empirical material and theory (Dubois and Gadde 2002).

To analyze the data, we first engaged in within-case and then cross-case analysis (Eisenhardt 1989). The within-case analysis included coding and analyzing each case separately. We analyzed the context factors of each case company and the resulting behaviors, cognitions, and affective states of the individual informants. To familiarize ourselves with the data, we engaged in open coding to identify first-order concepts. Simultaneously, we iterated between data collection and open coding (Eisenhardt 1989). We used the software MAXQDA for coding. Eventually, we derived 55 open codes describing what characteristics process mining use patterns can take on for individual users. One author conducted the coding in MAXQDA and the coding results were extensively discussed with the other authors in bi-weekly meetings. We also used these meetings to discuss differing opinions about the coding and, later, the aggregation of data until we reached consensus. Table 4 exhibits examples of open codes and respective illustrative interview data.

Then, in line with the abductive approach (Dubois and Gadde 2002), we integrated the theoretical notion of IS use comprising the three dimensions of behavior, cognition, and affect (Ortíz de Guinea and Webster 2013) with the emerging open codes. To find similarities and differences across cases, we grouped similar configurations of behavior, cognition, and affect characteristics. This resulted in distinct spectrums of use characteristics for quality and quantity in use, the level of cognitive effort, as well as the outcome of expectation-confirmation, and positive and negative affective states. We assigned those spectrums of use characteristics into the three overarching use

Table 4 Open coding examples of the use pattern dimensions

Illustrative data	Open code	Use dimension
<i>"I do not really analyze many processes myself. At the moment, we are responsible for establishing a CoE for the topic and anchoring the topic better in the Group. For this reason, I have more of an organizational role in setting up the system landscape and establishing governance structures accordingly." (MediCo1)</i>	Non-routine task	Behavior
<i>"I started in the business intelligence environment [...]. But then I saw ProMi and thought it could also look like that. That impressed me a lot." (TeleCo1)</i>	Expectations typically exceeded	Cognition
<i>"I do not think I was shocked, sometimes it was a bit of a pity. But I mean, now it is being used, but it has only come in the last six months. I think it is sometimes hard to change the mindset there. Or it takes time." (SkyCo2)</i>	Frustration	Affect

Table 5 Coding examples of the spectrum of characteristics for use quantity

Illustrative data	Characteristics	Use dimension
<i>"They all have access, but it is not used by everyone." (AutoCo3)</i>	Non-use	Behavior
<i>"It is something I look at maybe once a week or once a month [...] I am really only a sporadic user of process mining at the moment. I would say even very, very occasionally only." (MediCo3)</i>	Irregular	
<i>"The monitoring with process mining we do every 14 days; the other report we do monthly." (SkyCo4)</i>	Regular	
<i>"My job revolves 100% around PM" (AutoCo1)</i>	Extensive	

dimensions of behavior, cognition, and affect. Table 5 provides an example of how we inductively derived one spectrum of characteristics. Particularly, we show exemplary quotes for the spectrum of characteristics of *quantity* belonging to the use dimension *behavior*.

Finally, we derived overarching dimensions depicting the user categories. Therefore, we clustered comparable typical configurations of use patterns, which resulted in the identification of four user categories: process mining influencers, power users, process participants, and strategic users. For example, we identified the process mining influencer pattern by grouping the following configuration: strategic steering tasks while using basic or intermediate process mining features on a regular or irregular basis. Hence, the cognitive effort required is low or medium, and the individuals categorized as process mining influencers report that their prior expectations are typically exceeded by their actual use. In addition to those behaviors and cognitions, affective states for process mining influencers are interest, motivation, and frustration about management's high expectations. We visualized the above-described analysis approach in a graphic that can be retrieved in Appendix B. Figure 1 exemplarily depicts the resulting data structure for the process mining influencer user category. The data structure is a graphical representation of how we arrived at the final user categories based on our

interview data. The data structures for the remaining user categories are visualized in Appendix C.

4 Findings

Our multiple case study provides insights into the evolution of process mining endeavors for each of the six case companies from initial adoption to the current state of post-adoption, as summarized in Table 6. With at least four years of process mining experience at the time of the interview conduction, all organizations are in the post-adoption phase with the process mining software of ProMi. All companies have undergone process mining-related developments, such as migrating from the on-premises to the cloud version and the continuous development of use cases and the respective user base.

By comparing the informants' characteristics across cases, we delineate four process mining user categories based on their distinct configurations of behaviors, cognitions, and affective states: (1) process mining influencers, (2) power users, (3) process participants, and (4) strategic users. This user categorization is not based on the formal job titles but on their actual process mining use patterns, meaning their typical spectrum of behaviors, cognitions, and affective states. We posit that these patterns are dynamic and depend on the use context or the antecedents

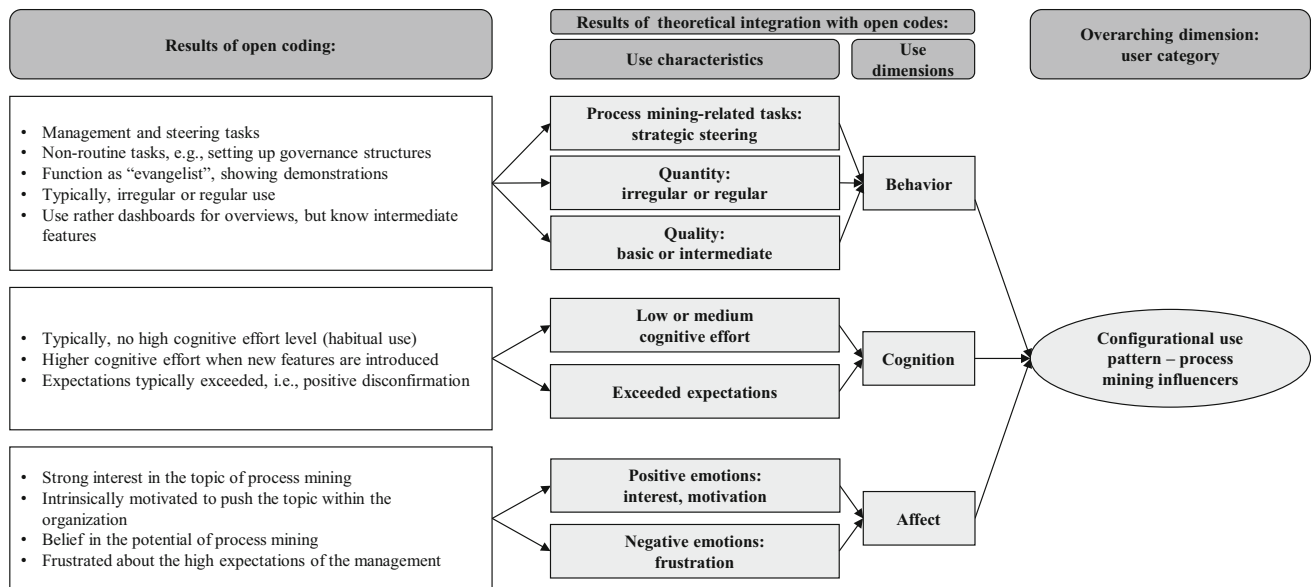


Fig. 1 Data structure example of process mining influencers

to individual process mining use. Hence, when describing the manifestations of the use dimensions per user category in the following, we also describe what context factors drive them. Importantly, as different context factors stimulate different use characteristics, the spectrum of characteristics per user category can be relatively broad and multifaceted.

We deliberately avoid assigning our informants to one category only, as we discovered that many of them transfer among categories over time, meaning they undergo user evolutions during their careers. For instance, informant AutoCo2 started using process mining as an operative shopfloor worker at AutoCo but can be categorized as a power user at the time of the data collection. Further, we do not formulate static or ideal type patterns for each user category but outline the spectrum of characteristics that behavior, cognition, and affect can take. In Table 7, we provide an overview of the four user categories’ configurations of process mining use characteristics, which we elaborate on in detail in this section.

In the following, we describe our findings on individual process mining use patterns per user category and what antecedent factors drive them. Thereby, we adhere to the following structure. First, we detail the behavioral use dimension of a specific user category, including the tasks they perform with process mining, the quantity and then the quality of use behavior. Second, we describe the expectation-confirmation relation and cognitive effort level. Third, we outline the typical spectrum of positive and negative affective states when using process mining.

4.1 Process Mining Influencers

4.1.1 Behavior

Our data analysis identified a category of users who act as visionaries who inspire other organizational stakeholders and get them enthusiastic about process mining. We refer to them as process mining influencers. “[Informant SkyCo1] was the pioneer who really fought with his department for the [process mining-related] topics to be implemented” (SkyCo5). We often observed that CoE leads, internal consultants or project managers take on the role of a process mining influencer. They view it as their task to disseminate process mining use within their organization. “I have a vision with which I want to contribute” (SkyCo1). While process mining influencers are well-versed in the features and functionalities of process mining, including intermediate features, they typically do not use process mining to conduct analyses or support operations. Instead, they engage with process mining through showcases or by loading play data. “I have my showcases, which I use to tour the organization and show them” (TeleCo1). This rather strategic user category focuses on non-routine tasks such as setting up organizational structures and winning resources for process mining undertakings, as well as managing and steering them. Consequently, management buy-in is pivotal for this user category as a positive antecedent to their use. This comprises the provision of resources, such as the funding for a CoE and the dedication of manpower to the topic of process mining by the top management. “What is important to us is that [...] there is a back-up from the top that says we want to do it this way.

Table 6 Within-case information

Key data	Summary of process mining evolution, governance mode, and current process mining endeavors
<u>TeleCo</u> Six years since process mining implementation (ProMi on-premises version) Process mining focus: Procurement	TeleCo introduced process mining in 2016 and installed a process mining CoE with 20 employees. TeleCo is occasionally supported by ProMi consultants, e.g., for the ongoing migration from the process mining on-premises to the cloud version. With the former, licensees are unlimited, which led to around 2,000 active process mining users. Today, ProMi is rolled out to more than 150 use cases. The focus of our interviews was especially on <i>Procurement</i> processes such as <i>Purchase-to-Cash</i> . At the interface between the businesses and the CoE are operational steering departments that manage the process mining analyses of their respective field. TeleCo focuses on combining process mining with automation solutions. Due to process mining, TeleCo realized substantial savings
<u>SkyCo</u> Four years since process mining implementation (ProMi cloud version) Process mining focus: Ground handling	In 2018, SkyCo adopted process mining and installed a CoE with six employees. Due to the COVID-19 pandemic, which hit the aviation industry severely, the process mining CoE headcount was reduced to one. The rising demand for process mining in the business departments cannot be served by the small CoE, and in departments where touchpoints with the process mining expert are lacking, process mining use sometimes stops completely. Where the CoE is active, process mining is further developed, and new features are employed. In our interviews, we mainly targeted process mining in relation to <i>Ground Handling</i> processes. SkyCo was able to reduce minutes of delay and thereby increase customer satisfaction significantly
<u>MediCo</u> Ten years since process mining implementation (ProMi cloud version) Process mining focus: Procurement	Having met the founder of ProMi in 2012 at a fair, MediCo's internal audit implemented process mining, and years later, other departments followed. So far, process mining is driven in a decentralized manner by process mining experts in the departments. Recently, the management decided to establish a central process mining and robotic process automation CoE, which is to be set up. Even though process mining licensees are not limited, most employees are non-users. Today, process mining undertakings at MediCo can be described as rather project-based initiatives for interested employees, and only small process adjustments have been realized so far. The interviews focused on <i>Procurement</i> processes such as <i>Purchase-to-Cash</i>
<u>AutoCo</u> Six years since process mining implementation (ProMi cloud version) Process mining focus: Manufacturing	Reacting to a rapidly changing environment, AutoCo implemented process mining in 2016. In 2018, a large process mining CoE was set up, divided into subject matter experts with clear responsibilities. Each department has a process mining Center of Competence with key users installed. These clear structures resulted in a fast centralization and professionalization of the process mining undertakings. Today, more than 40 processes within AutoCo are evaluated with process mining by more than 850 users. Our focus within the data collection was on the <i>Manufacturing</i> processes. However, not everyone is and wants to become a user due to a lack of IT-affinity and system fatigue. AutoCo has a strategic long-term partnership with ProMi
<u>CongloCoA</u> Eleven years since process mining implementation (ProMi on-premises version) Process mining focus: Procurement & Sales	Since the process mining implementation in 2011, CongloCo and ProMi have been strategic partners. In 2015, CongloCo installed a central process mining CoE, which consists of about 30 employees today. A comprehensive project-based rollout of process mining within CongloCo's global divisions followed in 2016. CongloCoA is currently migrating its process mining on-premises landscape to the cloud. In total, more than 30 processes are analyzed with process mining, and the process mining user base has more than 6,000 active users. Depending on the business department, either Centers of Competence or internal consultants provide the departments with process mining competence. We focused on <i>Procurement</i> processes such as <i>Purchase-to-Cash</i> and <i>Sales</i> processes such as <i>Order-to-Cash</i>
<u>CongloCoB</u> Four years since process mining implementation (ProMi & VeTwo on-premises version) Process mining focus: Procurement & Sales	In 2018, CongloCo carved out CongloCoB, which is still majority-owned by CongloCoA. CongloCoB has a service contract with CongloCoA to share the use of the latter's process mining platform. In addition to ProMi's process mining solution, CongloCoB applies the business intelligence-based process mining tool VeTwo. So far, CongloCoB does not have its own central team to coordinate its process mining initiatives, but it is planned to set up a CoE soon. Today, process mining initiatives happen rather in silos driven by experts in the business departments. At CongloCoB, there are about 200 process mining users. We focused on <i>Procurement</i> processes such as <i>Purchase-to-Cash</i> and <i>Sales</i> processes such as <i>Order-to-Cash</i>

Table 7 Overview of process mining user categories' use patterns

		Process mining influencers	Power users	Process participants	Strategic users
Behavior	<i>Main tasks</i>	Strategic steering of process mining within the organization	Great variety of process mining-related tasks	Process performance monitoring	Decision-making about process changes
	<i>Quantity</i>	Irregular or regular	Regular or extensive	Non-use to regular	Non-use to regular
	<i>Quality</i>	Basic or intermediate features	Intermediate or advanced features	Non-use or basic features	Non-use or basic features
Cognition	<i>Cognitive effort</i>	Low or medium	Medium or high	No to medium, strong habitualization	No to medium, strong habitualization
	<i>Expectation-confirmation</i>	Exceeded	Undercut or fulfilled	Fulfilled or exceeded	Undercut or exceeded
Affect	<i>Negative</i>	Frustration	Frustration	Work-related fears	Status-related fears, disappointment
	<i>Positive</i>	Interest, motivation	Interest, motivation, enthusiasm	Interest, motivation	Interest, motivation, gratefulness

We believe in it. And that the management defends the tool" (TeleCo1). To fulfill their tasks, process mining influencers primarily use the basic process mining features and focus on the dashboard function. "I rarely descended into the process explorer or into the variants" (SkyCo1).

4.1.2 Cognition

Regarding cognition, we found that process mining influencers are positively surprised about what they can achieve with process mining in their organization. Therefore, their previous expectations are typically exceeded. "That really impressed me, that potential of how analyses can look and how well they can be structured" (TeleCo1). Process mining influencers, who have a solid understanding of how process mining works but primarily use its basic features, typically exert minimal cognitive effort. As one informant noted, they tend to "use it more on the side" (SkyCo1). However, reflective thinking increases when they engage with new features that they can present to organizational stakeholders.

4.1.3 Affect

Our multiple case study revealed that process mining use is generally an emotional matter, and no user category is indifferent towards process mining. "It is very, very emotional. Dashboards are very emotional; we have often fallen on our face here" (CongloCoA2). Process mining influencers usually have very positive emotions toward process mining. They are strongly interested in the topic, believe in the potential provided by process mining, and are highly motivated to develop process mining within their organizations. "I think I am the biggest advocate of

this topic within the entire SkyCo" (SkyCo2). Process mining influencers' role as visionaries often entails greater visibility within their organization. This allows them to broaden their network and positively influences their career trajectory, such as through internal promotions. "I see it as an opportunity for me. So far, I have made good use of it" (TeleCo1). Yet, process mining influencers can sometimes feel frustrated, as they need to relativize unrealistic expectations that especially strategic users hold. This not only illustrates how important expectation management and educational work are as antecedents but also is a good example of how the use dimensions are interrelated: strategic users' cognitions (i.e., high expectations) lead to an affective response among process mining influencers (i.e., their frustration about unrealistic expectations they need to deal with). Figure 2 depicts the configurational use pattern of process mining influencers.

4.2 Power Users

4.2.1 Behavior

Unlike process mining influencers, power users conduct hands-on process mining by applying various advanced features in their day-to-day tasks. Power users engage with process mining regularly, often extensively, sometimes even full-time. Their behavioral system interaction demands a strong IT background and deep technological expertise. "My background maybe helped me because I used to work in IT" (CongloCoA5). Power users perform a wide range of tasks, including building dashboards (front-end), conducting process mining analyses, and sometimes taking care of backend-related tasks, such as preparing and loading data. In addition, they often advise on how to

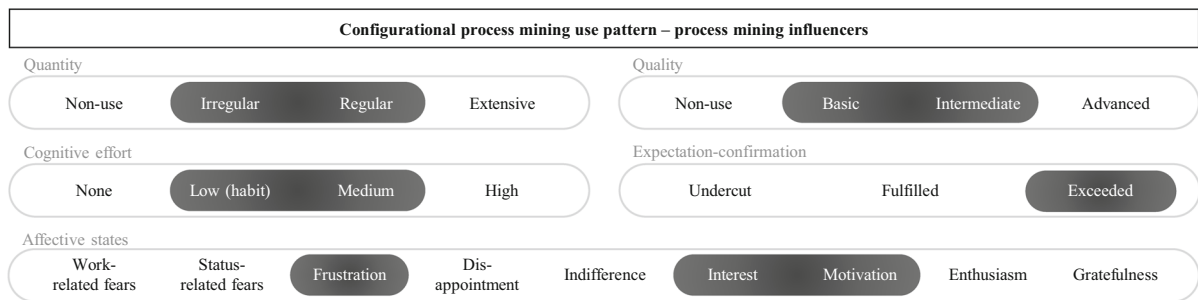


Fig. 2 Configurational use pattern of process mining influencers

change the process, explore new features and use cases, and engage strongly in the further technical development of process mining. *“The whole range from creating transparency to translating this transparency into measures and following them up in the long-term to ensure they are implemented. That is my daily business”* (CongloCoA4). Remarkably, when fewer people are dedicated to process mining within an organization, the lines between process mining influencers and power users get blurred. For example, informant SkyCo2 covers both strategic and operative process mining tasks. *“I cannot completely get away from the operations. [...] But mainly it is about developing the whole thing strategically”* (SkyCo2).

4.2.2 Cognition

The cognitive processes of power users, whether confirming or challenging their prior expectations, are shaped by their individual backgrounds and experiences. Our findings reveal that power users with professional experience in the IT field tend to have realistic expectations of process mining, resulting in fulfilled expectations. However, if power users lack deep technical expertise, they have higher expectations regarding the impact of process mining. As a result, their initial lofty expectations cannot be met, resulting in negatively disconfirmed expectations. *“I thought we could do more with it”* (CongloCoA4). Due to their extensive behavioral engagement, power users do not need to think actively when applying standard functionalities. However, using advanced or novel features or conducting non-routine tasks such as exploring new use cases is cognitively exhausting. *“Mostly, I use it quite naturally and automatically. And then there are these special cases where you really need to think”* (CongloCoA1).

4.2.3 Affect

Compared to process mining influencers, the positive affective state of power users towards process mining is even stronger. They are not only interested in process mining and are motivated to use and further explore it, but

they have fun doing so and are enthusiastic about it. *“I find it very, very fascinating”* (TeleCo3). They view process mining as enriching their daily work. *“I have the great pleasure of being in charge of this topic for production and logistics from the IT side, which I personally really enjoy”* (AutoCo1). Some power users even spend their leisure time further developing their expertise in process mining. *“I read the book of Wil van der Aalst. And I am following him all the time because he shares really good knowledge”* (MediCo2). Besides such positive feelings, power users also feel frustration when other users do not fully exploit the potential of process mining, especially when they have limited possibilities to address this issue. *“You can only empower them so they can identify process inefficiencies. But whether they then address them and change something or whether they just put them on the shelf and dust them off now and then, unfortunately, that is out of my hands”* (AutoCo1). In addition, continued resistance from others towards process mining use can be annoying and frustrating for power users. *“This [data privacy] is a huge, huge discussion. It is absolutely understandable, but it sometimes frustrates you”* (TeleCo2). They feel like they *“always have to unpack the same arguments”* (AutoCo4). Due to continuous bottom-up privacy concerns, they cannot implement some of their ideas. Therefore, *“buy-in from the managers”* (TeleCo1) is particularly important for the undertakings of power users. What can also be dissatisfying for power users is when their managers fail to appreciate their efforts in driving the development of process mining. *“It is a lot of work, and it is a kind of work which is not valued”* (MediCo2). The drive and motivation of power users can sometimes be dampened by the inertia of large organizations. *“You always have the feeling, especially in large corporations, that everything is progressing so incredibly slowly [...] That is sometimes frustrating”* (TeleCo2). Figure 3 summarizes the configurational use pattern of power users.

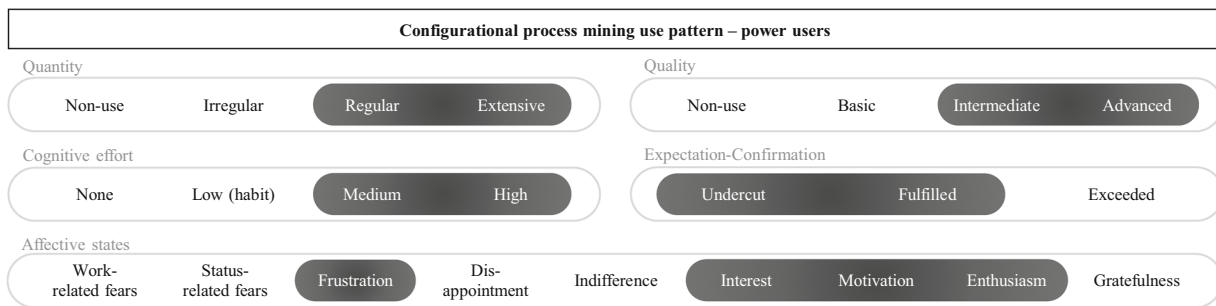


Fig. 3 Configurational use pattern of power users

4.3 Process Participants

4.3.1 Behavior

Process participants engage in operative process activities such as placing orders or paying invoices as part of their daily tasks. They typically use process mining to monitor the performance of the processes in which they are involved. Notably, process mining is an additional task rather than the focus of their role. Consequently, individuals in this category might decide not to use it at all or only to a minimum extent. *“I used to spend one hour per day with process mining on average. That was in the beginning. I was very curious, you know, trying to understand. [...] Now I use it not that much”* (CongloCoA5). Additionally, lacking digital maturity can contribute to non-use. For instance, when the degree of process digitalization is low and many process activities are manual, it is more difficult to employ process mining. *“The first step would be to digitize these processes so that they can be recorded with time stamps”* (SkyCo5). To enable the use of process participants, training and ongoing support by process mining experts such as the power users are needed. *“Definitely you need trainings, preferably a variety of things”* (SkyCo2). Most process participants we interviewed use process mining regularly but restrict their use to process monitoring, i.e., only basic features such as dashboards are applied. *“It is always these standard queries we make”* (SkyCo4).

As explained, this user category might not always actively use process mining tools, but they enable process analysis through their data contribution. Yet, it is especially crucial to consider this user category in process mining endeavors and support their process mining use, as management buy-in alone is insufficient. Effective process mining use in organizations is strengthened by the simultaneous support for process mining from the process participants. *“It is exactly this mixture between push [management support] and pull [operative process mining] that is the magic of it [...] because top management alone does not work in the long run”* (CongloCoA1). Their

experience with the operative process performance is important for the design and employment of process mining within their organization. *“We cannot do that without the operation team because they are the ones that know the process, and then based on the information they give to us, then we can understand”* (CongloCoA7). Such close collaboration between process mining experts and process participants requires a culture of trust and cooperation. *“I worked in operations for many years. [...] The people in the operations, you know, I understand them because I was working with them”* (TeleCo3).

4.3.2 Cognition

Process participants who actively use the software are typically very impressed by the value process mining can provide. Therefore, their previous expectations are positively disconfirmed. *“I think it is a really good tool, compared to what we had before”* (SkyCo4). However, the positive confirmation is curtailed if technical issues persist. *“We still need to get it up and running, so the expectations are, so far, not really exceeded. But I do not think I can attribute it to the tool”* (MediCo3). The cognitive effort level depends on the use behavior. Logically, non-users cannot show any active reflections. From all user categories, process participants and strategic users have habitualized process mining use the strongest. They made process monitoring with process mining dashboards a habit, e.g., through regular formal performance meetings. *“The application that we have works without you having actually to think about it”* (SkyCo4).

4.3.3 Affect

Impressed by the capabilities of process mining by which unprecedented transparency can be generated, most process participants are interested in and motivated to use the tool. *“That you can follow the process, that is very exciting for most people”* (CongloCoA3). Through process mining, the operative staff receives access to previously unavailable knowledge. *“Now they have a better overview about what*

is going on” (TeleCo3). This leads to increased empowerment of process participants. Also, the transparency of process mining can be used to improve the processes so that mindless manual tasks can be omitted. As a result, capacities are freed that can be shifted to more value-adding and enriching tasks. *“The colleagues no longer had to worry about orders moving from one status to the next and having to rework them, and so on. Instead, they were able to focus more on core purchasing activities, such as selecting suppliers, renegotiating, taking discounts, etc.”* (TeleCo1). This can upgrade an individual’s job profile and increase job satisfaction. *“We are just getting rid of the routine. [...] To make life simpler and easier for employees and raise their employment satisfaction”* (MediCo2). However, the feeling of motivation to use process mining depends on the individual’s background. We noticed that individuals with an IT background and technology affinity are typically more motivated to use process mining.

We observed that individuals’ mindsets, as well as the organization’s culture regarding the handling of process mining-induced transparency, can positively or negatively influence how process participants feel about process mining. *“It depends on whether they use it [the transparency] to punish people or to motivate people”* (AutoCo5). Process participants particularly fear that their individual behavior is made transparent and traced by process mining. Many informants mentioned that *“people are scared of performance tracking”* (ProMi1). These fears are mitigated when the organization provides strict data anonymization policies, frequent communication, and well-expounded change management. If process participants trust the anonymity of their data, they are more willing to use process mining. *“Theoretically, you could make employees super transparent. That is simply strictly forbidden. [...] If every click would be tracked, then people would not want to use ProMi anymore”* (TeleCo2). Furthermore, the data suggested that it is a positive antecedent not to limit licensees so all users can verify what data is displayed in process mining. *“When we do limit licensees to just the managers, that is when your customers [the organizations that employ process mining] start to struggle”* (ProMi1). Fear often plays a role for process participants, be it the substantial fear of losing the job – *“What happens to my job if the process is eliminated?”* (CongloCoA3) – or other work-related fears. For instance, process participants fear losing freedom in performing the process according to their own preferences or increased performance pressure. *“Perhaps somehow more performance will be demanded of those involved in the process”* (AutoCo1). Also, process participants fear that process mining will increase their workload, as process mining is *“another tool that is certainly more work if I then have to*

operate or maintain it” (SkyCo5). We visualize the configurational use pattern of process participants in Figure 4.

4.4 Strategic Users

4.4.1 Behavior

Interestingly, process participants and strategic users reveal similar behavioral patterns but differ in use purpose, i.e., the tasks they perform with process mining. Strategic users typically cover managerial roles and are responsible for the processes. Hence, they apply process mining to monitor the processes they oversee and initiate process improvements. *“The purchasing manager can see the key figures every day. He can see how efficient his employees are and can address this accordingly. Look here, I noticed that the KPI [key performance indicator] is going down”* (TeleCo1). Thus, individuals with decision power over processes have an important role in steering whether process mining brings business value or not. *“The ones [processes] for which the managers are taking ownership are the ones that are successful right now. There are some use cases that I have implemented that do not realize a lot of value, and it is because the managers have not made it their personal responsibility”* (ProMi1). For that reason, strategic users must be convinced to use process mining and act upon the insights generated. The quantity of use by strategic users ranges from regular to none. While most use process mining regularly, some individuals in the strategic users category refuse to use process mining, often caused by a lack of awareness or training. The provision of sufficient training resources is pivotal in developing the capabilities of strategic users. To ensure user enablement, a centralized process mining governance mode, in the form of a CoE, appears to be beneficial. This way, users know exactly where to seek support and guidance. *“A recipe for success is definitely to centralize the topic”* (CongloCoA1). Also, the technological context can be set up in a way that enables process mining use for strategic users, e.g., by high levels of data quality and user-friendliness. *“It is important that the frontend you build is useful, that dashboards and tools are easy to use for the end users and really make a difference”* (AutoCo5). Yet, many strategic users only want to consume reports but not engage directly with the process mining software. *“The head of the department receives the evaluation by e-mail”* (AutoCo3). To fulfill their process responsibility with process mining, we find that most strategic users apply basic features such as the dashboard function.

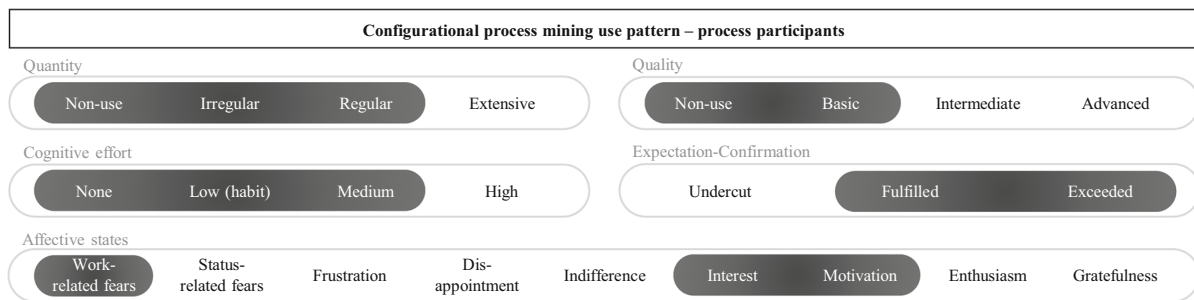


Fig. 4 Configurational use pattern of process participants

4.4.2 Cognition

Interestingly, strategic users exhibit great variability in terms of their cognitions depending on their particular context and individual mindset. When compared to their pre-use expectations, strategic users usually mention positive disconfirmation, i.e., the previously held expectations were exceeded during the use of process mining. They view the technology as a “milestone” (SkyCo4) that has the power to change something and “is not the umpteenth system that promises you that everything will be better, but, in reality, it is just another system” (TeleCo2). However, in the absence of sufficient expectation management, strategic users tend to hold unrealistic expectations that cannot be fulfilled, i.e., they experience negative disconfirmation. “They simply expect that you throw in some data, and a process falls out. Experience has shown that this cannot be fulfilled in most cases [...] These are really management expectations. They have seen a presentation on a topic and simply expect a lot to work out of the box” (MediCo1). Like process participants, strategic users have habitualized process mining use strongly, as they made process monitoring with process mining dashboards a habit, e.g., through regular formal performance meetings. “We have weekly performance meetings based on the various dashboards, and in the course of that, we dig into specific issues” (SkyCo3). As an exception, those strategic users who want to contribute to the further development of the process mining tools, together with power users in their company, need to invest more cognitive effort. “Two members from my team [...] look at the performance every day and develop it. And the three of us are also developing a new feature” (SkyCo3).

4.4.3 Affect

As strategic users are “interested in making the process better” (AutoCo2), “for the most part, managers are grateful” (ProMi1) that process mining aids them in achieving greater process excellence. Some strategic users are interested in process mining and are motivated to apply

the software, while other strategic users feel frightened as they fear losing their expert status. “Process mining is often not well tolerated because people suddenly had more knowledge than other people with 20, 30 years of experience. [...] And the second thing is the perceived loss of face. [...] It is just such a power issue” (CongloCoA2). However, given a culture of cooperation and trust, strategic users appreciate the empowerment of process participants. “Transparency means that specific knowledge is not restricted to individual people but becomes collective property. [...] I do not feel threatened” (SkyCo3). Some strategic users refuse to apply process mining as they do not want to see the full extent of inefficiencies. There is a “fear of transparency, to see in black and white what is really happening” (CongloCoA2). Thus, transparency requires cultural readiness characterized by error tolerance and the willingness to change. “It is a cultural issue. If I am in a company that is very data-driven, KPI-driven, then I would expect less resistance there” (SkyCo1). Building on the elaboration above, some strategic users have unrealistic expectations of process mining that “quickly disillusion” (SkyCo2) when actively using process mining. To prevent such disappointment, strategic users need to understand that process mining itself does not fix any process but is “the beginning of a long journey” (CongloCoA2). Figure 5 exhibits the configurational use pattern of strategic users.

5 Discussion

Process mining is an emerging technology that has experienced a rapid increase in adoption across industries, alongside a significant rise in the market capitalization of its vendors (vom Brocke et al. 2021). Its application transforms organizational business process analysis by enhancing visualization and optimization. It offers significant business benefits, including increased process efficiency, financial gains, and non-monetary advantages, such as improved process compliance and heightened customer satisfaction (Badakhshan et al. 2022). An emerging stream of research has accumulated on the organizational

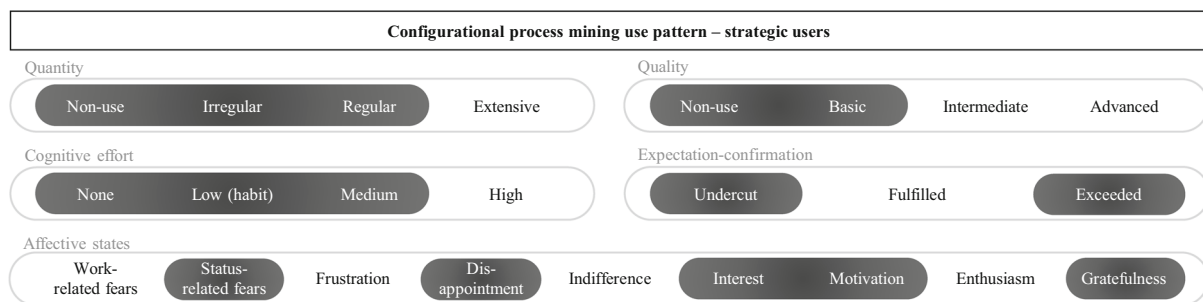


Fig. 5 Configurational use pattern of strategic users

adoption and application of process mining technologies (e.g., Eggers et al. 2021), to which this study contributes. So far, studies focus on initial opportunities and challenges associated with process mining adoption (Martin et al. 2021; Grisold et al. 2020) or outcomes of its application (Badakhshan et al. 2022). The transition from adoption to value within organizations remains largely unexplored. Therefore, we conducted an empirical investigation into individual-level use, which is the critical link between organizational adoption and the realization of tangible value (Devaraj and Kohli 2003).

Our study makes two significant contributions to the research on individual process mining use in organizations, responding to calls for empirical examination of this topic (vom Brocke et al. 2021). First, we identify distinct configurations of process mining use patterns, which allow us to propose an empirically grounded user categorization, distinguishing users as process mining influencers, power users, process participants, and strategic users. Second, we explore the behaviors, cognitions, and affective states of users from each category and the antecedents that shape these use patterns. The following section elaborates on our theoretical contributions, discusses our findings in relation to existing research, and outlines practical implications, limitations, and directions for future studies.

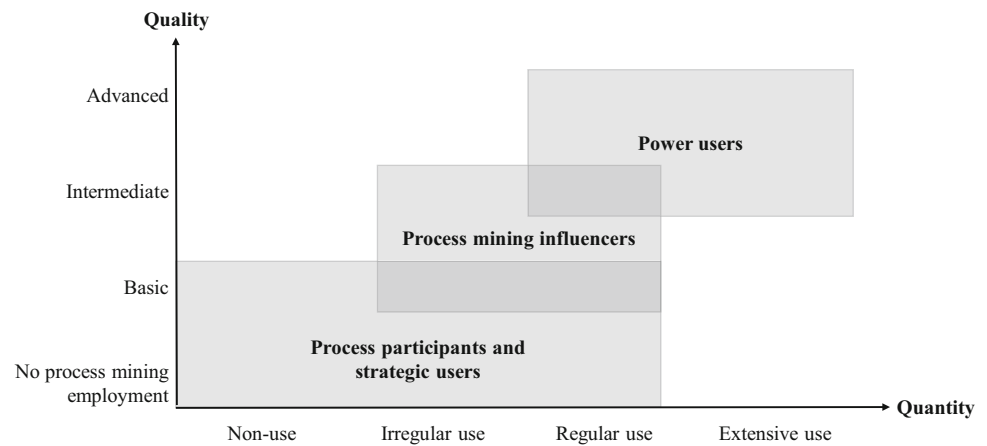
5.1 A Process Mining User Categorization

Individual process mining use differs concerning behavior, cognition, and affect. We identified four distinct use categories based on their use pattern configuration: process mining influencers, power users, process participants, and strategic users. Our study contributes to research on process mining use in organizations (Grisold et al. 2020) and user categories (vom Brocke et al. 2021). So far, the literature has conceptually described how established business process management roles (process analysts, process participants, system engineers, and process stakeholders) perceive or use process mining (vom Brocke et al. 2021). According to these studies, it is mainly the process analysts who work with process mining (Zerbato et al. 2022),

whereas system engineers are responsible for providing data (vom Brocke et al. 2021). This implies that other organizational roles are rather passively interacting with process mining. Previous research does not question that the individual use of process mining might lead to new user categories that are not based on their roles but on their cognition, affect, and behavior toward process mining. Our study empirically extends the descriptions of previously mentioned groups and consolidates them in a comprehensive categorization based on process mining users' behavior, cognition, and affect (Ortíz de Guinea and Webster 2013). In addition, we extend prior user categorizations (vom Brocke et al. 2021) by introducing process mining influencers as a pivotal user group, enhancing the effectiveness of process mining initiatives.

Figure 6 provides an overview of our user categories focusing on the behavioral dimension. The figure visualizes individuals' behavioral interaction with process mining tools along the axes of quantity (ranging from non-use to extensive use) and quality (ranging from no process mining employment to advanced feature use). The figure highlights two key insights. First, power users can be identified as a distinct user category based on their behavioral interactions. However, the behavioral patterns of process mining influencers overlap significantly with those of process participants and strategic users, making these groups less distinguishable in this dimension. Second, the figure illustrates the spectrum of both quantitative and qualitative interaction within a user group. Thus, while this figure summarizes the user categories in terms of behavior, we want to emphasize the relevance of going beyond the behavioral dimension to paint a more nuanced picture of process mining use in organizations.

In the following, we discuss the user categories. First, our multiple case study shows that **power users** extensively employ process mining and deeply understand the full feature range. Previous studies have referred to this user category as process analysts (vom Brocke et al. 2021; Zerbato et al. 2022). Power users typically even cover backend-related tasks such as data preparation and connection. That is especially true when the power users are

Fig. 6 User categories based on their behavioral interaction

part of process mining CoEs. In contrast, vom Brocke et al. (2021) assign those activities to another category, i.e., system engineers. We argue that the separation into system engineers and process analysts does not adequately reflect individuals' interaction with process mining in practice, as they operate similarly. Based on our data, we cannot distinguish these two user categories. Further, while those studies describe power users based on their behaviors in using process mining, we extend the behavioral component with affective and cognitive aspects. Our findings portray power users' enthusiasm towards process mining but frustration about resistance from other actors and regulations, hindering their process mining-related ambitions within the organizations.

Second, previous research has conceptually categorized managers as process stakeholders (vom Brocke et al. 2021). Our findings correspond with previous studies stating that managers usually function as agenda setters for business process analyses and improvements (Grisold et al. 2020; vom Brocke et al. 2021). Hence, we refer to them as **strategic users** of process mining. Nevertheless, previous scholars' descriptions imply that strategic users do not actively interact with process mining. Indeed, we found that strategic users mainly use the dashboards provided within process mining tools to make decisions, but we observed that some strategic users conduct process mining themselves hands-on. For instance, informant SkyCo3, overseeing ground handling processes at SkyCo, closely works together with their CoE to develop new features and conducts process analyses with process mining regularly. Beyond the behavioral aspects that we deepened in our study, our data suggests that strategic users see process mining as a valuable tool that supports them in improving processes under their responsibility, while others fear knowledge democratization and the loss of their expert status.

Third, **process participants** were previously solely described as passive data generators (vom Brocke et al.

2021). However, we found that some process participants actively use process mining to monitor the performance of their process. While there are indeed many passive data contributors (i.e., non-users) among the process participants, others employ basic features on a regular basis. Beyond behavior, work-related fears are often present among the process participants (Stertz et al. 2021), such as the fear of being traced or losing their job due to increased efficiency. However, process mining can also be positively perceived, as it can, for example, free process participants from tedious manual tasks.

Finally, we add to previous user categories by introducing the category of **process mining influencers** who inspire and motivate organizational stakeholders. While this category has not yet been described in process mining research, it is highly influential for successful organizational value creation from process mining. Process mining influencers are responsible for developing the potential of process mining in the organization and not for conducting operative analyses. Our data shows that process mining influencers believe in the value of process mining and are enthusiastic about it. The process mining influencers' characterization shares some commonalities with *digital evangelists*, a type of the C-level position of Chief Digital Officer (Singh and Hess 2017). Like digital evangelists, process mining influencers are visionaries and push this digital topic within the organization. This function seems especially relevant when convincing large and fragmented organizations of this specific digital technology and fostering its diffusion.

5.2 Dimensions of Process Mining Use Patterns and Influencing Factors

Our second contribution refers to the detailed and integrated description of process mining use encompassing behavioral, cognitive, and affective aspects and antecedents that drive the configuration of use patterns. While

most studies investigate single dimensions, we extend the line of work that considers use beyond behavioral and “lean” measurements in the post-adoption context (Ortíz de Guinea and Webster 2013; Bagayogo et al. 2014). Here, we detail the interplay of different IS use dimensions and their antecedents. Numerous impact factors unveiled in our data are in line with antecedents from adoption literature on conventional business intelligence tools (e.g., Ain et al. 2019) as well as other big data analytics solutions such as predictive analytics (e.g., Schoenherr and Speier-Pero 2015) and data mining tools (e.g., Huang et al. 2012). This implies that relevant initial adoption factors maintain relevance in the post-adoption stage. Regarding antecedents, our findings highlight the importance of contextual factors like data policies, bottom-up forces, and process mining centralization for shaping individual process mining use in practice. In terms of use dimensions, our findings emphasize the unique role of affect for process mining use.

First, in line with Stein et al. (2015), we discovered that affect refers to a range of emotions and not a single affective state. Our findings portray mixed and even contradicting emotions when using process mining. Positive affective states often appear with extensive process mining use behavior and a wide range of feature employment. While previous scholars also pointed out negative feelings in IS use, such as anxiety (Brown et al. 2004), process mining is unique as existential fears can arise within the negative emotional facet. These fears are not only a phenomenon for initial adoption but persist even years after implementation. Process mining is particularly interesting for studying emotions in a work context, as it creates unprecedented transparency that promotes, e.g., privacy concerns among individual workers (Grisold et al. 2020). Even though there might not be behavioral use components at play for every employee, i.e., there are non-users, individuals still interact with systems through cognitive and affective dimensions with potentially far-reaching consequences.

Second, our analysis highlights how organizations treat personal data can influence process mining use. Based on our interviews, we conclude that the fear of being traced is deeply rooted and, therefore, should be considered by companies. Our findings align with Martin et al. (2021), who note that individual performance tracing might lead to defensive behavior or the feeling of being controlled. To protect personal data, scholars suggest privacy guidelines (Mannhardt et al. 2018) or privacy-preserving technical designs for process mining that are compliant with the European General Data Protection Regulation (Mannhardt et al. 2019). To encounter data privacy concerns, scholars have recommended that organizations should carefully determine the purposes of data use (van den Broek and van Veenstra 2015), offer individuals control over the data

collected on them (Greenaway et al. 2015), and adopt strict data policies (Tallon et al. 2013). Our case organizations have such policies in place and strictly enforce them, at least in Germany. In part, the informants reported that individual performance tracking is common practice in branches located in other countries. Moreover, previous studies show that trust is a strong driver for process mining use (Lohoff and Engert 2024). Hence, trust must be considered in personal data protection policies to avoid negative affective states among process mining users.

Interestingly, some researchers posit that using process mining for individual-level evaluation positively affects individuals, e.g., through fairer balancing of workload (Pika et al. 2017) and objective benchmarking of teams, including the guidance of low-performing and rewarding high-performance teams (Aysolmaz et al. 2021). However, it is important to note that performance might also consist of other dimensions that cannot be displayed with process mining, such as the complexity of a particular task. Overall, the findings call for placing users and their affective states central to process mining undertakings of organizations.

Third, scholars stress that management commitment remains relevant beyond the initial adoption of big data analytics technology (Gunasekaran et al. 2017; Sodero et al. 2019). Likewise, our informants mention continued management buy-in as critical. However, we also find that bottom-up pull, i.e., support from non-managers for process mining, gains importance in the post-adoption phase. While current research does not stress bottom-up pull as an enabling antecedent, we posit that, ideally, both forces are in place simultaneously: bottom-up pull and top-down push (Baiyere et al. 2020). This underscores the relevance of all individuals interacting with process mining. Process participants need to engage with process mining for data contribution, process mining influencers continuously need to increase awareness across the organization, power users need to prepare data and set the technical frame for process analysis, and strategic users, being decision-makers, need to use outputs for effective process execution. Here, management must signal priority and enable process mining use by providing technical access, training, time resources, and relevant incentives. Yet, top-down imposition is insufficient to leverage the value of process mining, as process participants need to engage with the system, and fears could block system exploitation.

Fourth, contemporary literature debates centralizing or decentralizing big data analytics capabilities (Günther et al. 2017), such as process mining capabilities. Our empirical evidence suggests that central coordination is beneficial for process mining activities, for instance, through a dedicated CoE (Reinkemeyer et al. 2022). Centralized process mining efforts increase organizational awareness and

positively influence cognition and affect of individual users. Decentralized process mining structures, in contrast, were too fragmented to provide the organization with sufficient leverage in our case companies.

Finally, we propose a notion of constant change underlying process mining use in practice related to properties of digital process innovations (Lohoff 2022). Process mining is currently developing rapidly (Kerremans et al. 2021). Even though being far beyond initial adoption, as our first case organization adopted process mining more than a decade ago, the cases show that implementation is still work-in-progress. Even the earliest adopters, MediCo and CongloCo, continue to adjust their process mining employment by adding additional features and technologies, e.g., task mining or robotic process automation. Research emphasizes studying post-adoption from the perspective of continued and habitualized use (Bagayogo et al. 2014). Our study suggests that current digital technologies are developing rapidly, whereby a state of habitualized and continued use is hard to reach. Thus, future post-adoption research should consider IS use from the perspective of continuous development and adjustment, where habitualized use might not necessarily be favorable.

5.3 Practical Implications

Based on insights from large organizations with substantial process mining experience, our study offers four main implications for leveraging the use of process mining in practice. First, this study helps managers identify different process mining user categories based on their unique configurations of behavior, cognition, and affect. While use behavior might be similar for some groups, cognition and affect set user categories apart. Our findings help practitioners understand how the users of these categories interact with process mining, and think and feel about it. Managers can build on this information to categorize employees regarding their interaction with process mining in order to ultimately optimize performance. For example, organizations can identify process mining influencers through their use pattern and provide them with more resources to share their knowledge and present showcases across the organization, allowing them to diffuse the technology. Moreover, as our study suggests that user categories are not stable, organizations can apply this categorization to develop individuals towards a particular category. For instance, through further training or specific task assignments, individuals could evolve from a process participant to a power user.

Second, our study highlights the underlying mechanisms and antecedents of use patterns. Underutilization can be an issue for process mining, as it is often applied like a business intelligence tool, i.e., it is not intuitive to users

what the unique and value-adding features of process mining are. This underutilization can be driven by limited cognition or negative emotion, which requires dedicated information and training as well as support from power users. We found that employees' fears regarding potential job terminations – even if irrational – can lead to limited application. Organizations should address this through effective communication. Our description of use patterns and underlying antecedents can support managers in shaping the process mining roadmap for their organization. Our findings call for a human-centered approach to process mining implementation and diffusion. Organizations should consider engaging and rewarding employees for actively using process mining rather than imposing it on them. This way, intrinsically and more extrinsically motivated users can be considered and motivated to use process mining in their unique ways.

Third, we argue that the use patterns are shaped differently depending on the composition of contextual factors, e.g., the existence of a process mining CoE. If the context is not enabling, neither beneficial use patterns nor favorable outcomes can result. We argue that practitioners need to create a flourishing environment to leverage the full potential of process mining and obtain a return on investment. According to our study, favorable antecedents are, for instance, dedicated process mining experts in a central CoE, training opportunities to build skills, and a culture of data policy transparency for a trusted atmosphere. These elements act as enablers for individual use in practice.

Fourth, beyond leveraging process mining use in organizations, our study offers implications for the enhancement of existing and the development of novel process mining techniques. Commercial software vendors and scholars pursuing technical research in the domain of process mining can use the derived user categories and their descriptions as a starting point for requirements engineering (Wagner et al. 2019). As our study shows, the users of process mining categories differ in their behavior, cognition, and affective states towards process mining software. These differences should be considered when developing new algorithms. For example, interviews with the strategic user category can be used to derive requirements for the development of algorithms that support decision-making about process changes while keeping the cognitive effort low. This will increase both the usefulness and the usability of process mining algorithms and corresponding software artifacts.

5.4 Limitations and Future Research

The limitations of our study pose avenues for future research. First, we used a qualitative approach to investigate process mining use in practice. Hence, our insights are

limited to the subjective experiences reported by the informants, including reports of users' quantitative use. We could not uncover use states outside the informants' awareness (Ortíz de Guinea and Webster 2013) and propose a quantitative approach to further investigate process mining use and its implications. As the context allows, future studies can rely on digital trace data (Grisold et al. 2024). Digital traces, such as online records of activities, would allow us to triangulate the complex reality of process mining use in practice.

Second, our study pioneers in providing the first empirically derived process mining user categorization and configurations of use patterns independent of organizational roles. However, we offer limited details on weak use or the failure to act on process mining suggestions. An in-depth investigation of resistance to conform to process mining suggestions (aversion behaviors) and non-use patterns in the context of process mining could be of special interest (Stein Dani et al. 2024; Lohoff and Engert 2024). To this end, future research can investigate unsuccessful adoptions of process mining and their contextual circumstances.

Third, our findings depict a snapshot of the informants' use patterns. Use patterns are dynamic and change over time due to continuous experience and learning (Bhattacharjee and Premkumar 2004). While we tried to cater to that with a spectrum of use patterns, we believe that longitudinal studies on the evolution of process mining use patterns are warranted.

Fourth, based on the topic of interest, we selected organizations that progressed in their process mining application beyond the state of initial adoption, leading us to a sample of large organizations. We acknowledge that IS use patterns and implications might look different for smaller organizations and vendors other than ProMi.

6 Conclusion

In light of the rapidly increasing relevance of process mining in academia and practice, this study is among the first to investigate individual-level process mining use beyond initial adoption. Based on a multiple case study, we empirically derived four user categories that differ concerning their use configurations comprising the interrelated use elements behavior, cognition, and affect: (1) process mining influencers, (2) power users, (3) process participants, and (4) strategic users. We describe the characteristics of each user category in detail. Our research adds to the body of knowledge on the use of process mining in organizations. Furthermore, our work is relevant for practice as the identified user categories can inform the development of new or the enhancement of existing

process mining algorithms. It can also help practitioners tailor measures to support use within the different user categories.

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References

- Agarwal J, Malhotra NK (2005) An integrated model of attitude and affect: theoretical foundation and an empirical investigation. *J Bus Res* 58(4):483–493
- Aguirre S, Parra C, Sepúlveda M (2017) Methodological proposal for process mining projects. *Int J Bus Integr Manag* 8(2):102–113
- Ain N, Vaia G, DeLone WH, Waheed M (2019) Two decades of research on business intelligence system adoption, utilization and success – a systematic literature review. *Decis Support Syst* 125:1–13
- Augusto A, Conforti R, Dumas M, La Rosa M, Maggi FM, Marrella A, Mecella M, Soo A (2018) Automated discovery of process models from event logs: review and benchmark. *IEEE Trans Knowl Data Eng* 31(4):686–705
- Aysolmaz B, Nemeth M, Iren D (2021) A method for objective performance benchmarking of teams with process mining and DEA. In: *Proceedings of the 29th European conference on information systems, Marrakesh*
- Badakhshan P, Wurm B, Grisold T, Geyer-Klingenberg J, Mendling J, vom Brocke J (2022) Creating business value with process mining. *J Strateg Inf Syst* 31(4):1–44
- Bagayogo FF, Lapointe L, Bassellier G (2014) Enhanced use of IT: a new perspective on post-adoption. *J Assoc Inf Syst* 15(7):361–387
- Baiyere A, Salmela H, Tapanainen T (2020) Digital transformation and the new logics of business process management. *Eur J Inf Syst* 29(3):238–259
- Beaudry A, Pinsonneault A (2010) The other side of acceptance: studying the direct and indirect effects of emotions on information technology use. *MIS Q* 34(4):689–710
- Bhattacharjee A, Premkumar G (2004) Understanding changes in belief and attitude toward information technology usage: a theoretical model and longitudinal test. *MIS Q* 28(2):229–254

- van den Broek T, van Veenstra AF (2015) Modes of governance in inter-organizational data collaborations. In: Proceedings of the 23rd European conference on information systems, Münster
- Brown SA, Fuller RM, Vician C (2004) Who's afraid of the virtual world? Anxiety and computer-mediated communication. *J Assoc Inf Syst* 5(2):79–107
- Burton-Jones A, Gallivan MJ (2007) Toward a deeper understanding of system usage in organizations: a multilevel perspective. *MIS Q* 31(4):657–679
- Burton-Jones A, Straub DW (2006) Reconceptualizing system usage: an approach and empirical test. *Inf Syst Res* 17(3):228–246
- Burton-Jones A, Stein M-K, Mishra A (2017) MIS quarterly research curation on IS use. *MIS Q Res Curations* 1(1):1–24
- Celonis (2022) Celonis raises \$1b in funding at \$13b valuation. <https://www.celonis.com/press/crunchbase-celonis-raises-1bil-ion>. Accessed 23 Oct 2024
- Chen H, Chiang RHL, Storey VC (2012) Business intelligence and analytics: from big data to big impact. *MIS Q* 36(4):1165–1188
- Chen DQ, Preston DS, Swink M (2015) How the use of big data analytics affects value creation in supply chain management. *J Manag Inf Syst* 32(4):4–39
- Cooper RB, Zmud RW (1990) Information technology implementation research: a technological diffusion approach. *Manag Sci* 36(2):123–139
- Davis FD, Bagozzi RP, Warshaw PR (1989) User acceptance of computer technology: a comparison of two theoretical models. *Manag Sci* 35(8):982–1003
- DeLone WH, McLean ER (2003) The DeLone and McLean model of information systems success: a ten-year update. *J Manag Inf Syst* 19(4):9–30
- Devaraj S, Kohli R (2003) Performance impacts of information technology: Is actual usage the missing link? *Manag Sci* 49(3):273–289
- Dubé L, Paré G (2003) Rigor in information systems positivist case research: current practices, trends, and recommendations. *MIS Q* 27(4):597–636
- Dubois A, Gadde L-E (2002) Systematic combining: an abductive approach to case research. *J Bus Res* 55(7):553–560
- Dumas M, La Rosa M, Mendling J, Reijers HA (2018) Fundamentals of business process management, 2nd edn. Springer, Heidelberg
- Eden R, Syed R, Leemans SJJ, Buijs JACM (2021) A case study of inconsistency in process mining use: implications for the theory of effective use. In: Polyvyanyy A et al (eds) Business process management LNCS, vol 12875. Springer, Cham, pp 363–379
- Eggers J, Hein A, Böhm M, Krcmar H (2021) No longer out of sight, no longer out of mind? How organizations engage with process mining-induced transparency to achieve increased process awareness. *Bus Inf Syst Eng* 63(5):491–510
- Eggers J, Hein A (2020) Turning big data into value: a literature review on business value realization from process mining. In: Proceedings of the 28th European conference on information systems, Marrakesh
- Eggers J, Häge M-C, Zimmermann S, Gewald H (2023) Assessing process mining use cases: a taxonomy of antecedents and value potentials of process mining. In: Proceedings of the 29th Americas conference on information systems, Panama City
- Eisenhardt KM (1989) Building theories from case study research. *Acad Manag Rev* 14(4):532–550
- El-Khatib R, Barki H (2012) The relationship between habit and intention in post-adoptive is acceptance. In: Proceedings of the 18th Americas conference on information systems, Seattle
- Fortune Business Insights (2024) Process mining software market, Report ID: FBI104792. <https://www.fortunebusinessinsights.com/process-mining-software-market-104792>. Accessed 17 Oct 2024
- Greenaway KE, Chan YE, Crossler RE (2015) Company information privacy orientation: a conceptual framework. *Inf Syst J* 25(6):579–606
- Grisold T, Mendling J, Otto M, vom Brocke J (2020) Adoption, use and management of process mining in practice. *Bus Proc Manag J* 27(2):369–387
- Grisold T, Kremser W, Mendling J, Recker J, vom Brocke J, Wurm B (2024) Generating impactful situated explanations through digital trace data. *J Inf Technol* 39(1):2–18
- Gunasekaran A, Papadopoulos T, Dubey R, Wamba SF, Childe SJ, Hazen B, Akter S (2017) Big data and predictive analytics for supply chain and organizational performance. *J Bus Res* 70:308–317
- Günther WA, Rezazade Mehrizi MH, Huysman M, Feldberg F (2017) Debating big data: a literature review on realizing value from big data. *J Strateg Inf Syst* 26(3):191–209
- Hsieh JJP-A, Wang W (2007) Explaining employees' extended use of complex information systems. *Eur J Inf Syst* 16(3):216–227
- Hsieh JJP-A, Rai A, Xu SX (2011) Extracting business value from IT: a sensemaking perspective of post-adoptive use. *Manag Sci* 57(11):2018–2039
- Huang TC-K, Liu C-C, Chang D-C (2012) An empirical investigation of factors influencing the adoption of data mining tools. *Int J Inf Manag* 32(3):257–270
- Igbaria M, Zinatelli N, Cragg P, Angele LMC (1997) Personal computing acceptance factors in small firms: a structural equation model. *MIS Q* 21(3):279–305
- Jain V, Kanungo S (2005) Beyond perceptions and usage: Impact of nature of information systems use on information system-enabled productivity. *Int J Hum-Comput Interact* 19(1):113–136
- Jasperson J, Carter PE, Zmud RW (2005) A comprehensive conceptualization of post-adoptive behaviors associated with information technology enabled work systems. *MIS Q* 29(3):525–557
- Kane GC, Labianca G (2011) IS avoidance in health-care groups: a multilevel investigation. *Inf Syst Res* 22(3):504–522
- Karahanna E, Straub DW, Chervany NL (1999) Information technology adoption across time: a cross-sectional comparison of pre-adoption and post-adoption beliefs. *MIS Q* 23(2):183–213
- Kerremans M, Srivastava T, Choudhary F (2021) Market guide for process mining. Gartner, Inc. <https://www.gartner.com/doc/reprints?id=1-289D1SO7&ct=211129&st=sb>. Accessed 27 Jan 2022
- Kim H-W, Chan HC, Chan YP (2007) A balanced thinking – feelings model of information systems continuance. *Int J Hum-Comput Stud* 65(6):511–525
- Lapointe L, Rivard S (2005) A multilevel model of resistance to information technology implementation. *MIS Q* 29(3):461–491
- Limayem M, Hirt SG, Cheung CMK (2007) How habit limits the predictive power of intention: the case of information systems continuance. *MIS Q* 31(4):705–737
- Lohoff L, Engert SP (2024) Better not invented here: Investigating algorithm appreciation in process mining. In: Proceedings of the 32nd European conference on information systems, Paphos
- Lohoff L (2022) A taxonomy for digital process innovation. In: Proceedings of the Pacific Asia conference on information systems, Taipei
- Mamudu A, Bandara W, Wynn MT, Leemans SJJ (2024) Process mining success factors and their interrelationships. *Bus Inf Syst Eng*. <https://doi.org/10.1007/s12599-024-00860-z>
- Mannhardt F, Koschmider A, Baracaldo N, Weidlich M, Michael J (2019) Privacy-preserving process mining. *Bus Inf Syst Eng* 61(5):595–614
- Mannhardt F, Petersen SA, Oliveira MF (2018) Privacy challenges for process mining in human-centered industrial environments. In: Proceedings of the 14th international conference on intelligent environments (IE), Rome

- Mans RR, Reijers H, Berends H, Bandara W, Prince R (2013) Business process mining success. In: Proceedings of the 21st European conference on information systems, Utrecht
- Martin N, Fischer DA, Kerpedzhiev GD, Goel K, Leemans SJJ, Röglinger M, van der Aalst WMP, Dumas M, La Rosa M, Wynn MT (2021) Opportunities and challenges for process mining in organizations: results of a Delphi study. *Bus Inf Syst Eng* 63(5):511–527
- Märušter L, van Beest NRTP (2009) Redesigning business processes: a methodology based on simulation and process mining techniques. *Knowl Inf Syst* 21(3):267–297
- Miles MB, Huberman AM (1994) *Qualitative data analysis: an expanded sourcebook*, 2nd edn. Sage, Thousand Oaks
- Miles MB, Huberman AM, Saldaña J (2014) *Qualitative data analysis: a methods sourcebook*, 3rd edn. Sage, Thousand Oaks
- Myers MD, Newman M (2007) The qualitative interview in IS research: examining the craft. *Inf Organ* 17(1):2–26
- Ortíz de Guinea A, Markus ML (2009) Why break the habit of a lifetime? Rethinking the roles of intention, habit, and emotion in continuing information technology use. *MIS Q* 33(3):433–444
- Ortíz de Guinea A, Webster J (2013) An investigation of information systems use patterns: technological events as triggers, the effect of time, and consequences for performance. *MIS Q* 37(4):1165–1188
- Pika A, Leyer M, Wynn MT, Fidge CJ, ter Hofstede AHM, van der Aalst WMP (2017) Mining resource profiles from event logs. *ACM Trans Manag Inf Syst* 8(1):1–30
- Reinkemeyer L, Grindemann P, Egli V, Röglinger M, Marcus L, Fabri L (2022) Accelerating business transformation with process mining centers of excellence (CoEs). Celonis SE, Fraunhofer Institute for Applied Information Technology FIT
- Schoenherr T, Speier-Pero C (2015) Data science, predictive analytics, and big data in supply chain management: current state and future potential. *J Bus Logist* 36(1):120–132
- Singh A, Hess T (2017) How chief digital officers promote the digital transformation of their companies. *MIS Q Exec* 16(1):1–17
- Sodero A, Jin YH, Barratt M (2019) The social process of big data and predictive analytics use for logistics and supply chain management. *Int J Phys Distrib Logist Manag* 49(7):706–726
- Stein M-K, Newell S, Wagner EL, Galliers RD (2015) Coping with information technology: mixed emotions, vacillation, and non-conforming use patterns. *MIS Q* 39(2):367–392
- Stein Dani V, Leopold H, van der Werf JME, Beerepoot I, Reijers HA (2024) From loss of interest to denial: a study on the terminators of process mining initiatives. In: International conference on advanced information systems engineering, Limassol
- Stertz F, Mangler J, Scheibel B, Rinderle-Ma S (2021) Expectations vs. experiences – process mining in small and medium sized manufacturing companies. In: Business process management forum. https://doi.org/10.1007/978-3-030-85440-9_12
- Straub D, del Giudice M (2012) Editor's comments: Use. *MIS Q* 36(4):iii–vii
- Subramani M (2004) How do suppliers benefit from information technology use in supply chain relationships? *MIS Q* 28(1):45–73
- Suddaby R (2006) From the editors: what grounded theory is not. *Acad Manag J* 49(4):633–642
- Syed R, Leemans SJJ, Eden R, Buijs JACM (2020) Process mining adoption. In: Fahland D et al (eds) Business process management forum. Springer, Cham, pp 229–245
- Tallon PP, Ramirez RV, Short JE (2013) The information artifact in IT governance: toward a theory of information governance. *J Manag Inf Syst* 30(3):141–177
- Thiede M, Fürstenau D, Barquet APB (2018) How is process mining technology used by organizations? A systematic literature review of empirical studies. *Bus Proc Manag J* 24(4):900–922
- Timmermans S, Tavory I (2012) Theory construction in qualitative research: from grounded theory to abductive analysis. *Sociol Theor* 30(3):167–186
- van der Aalst WMP, Adriansyah A, de Medeiros AKA, Arcieri F, Baier T, Blickle T, Bose JC, van den Brand P, Brandtjen R, Buijs J, Burattin A, Carmona J, Castellanos M, Claes J, Cook J, Costantini N, Curbera F, Damiani E, de Leoni M, Delias P, van Dongen BF, Dumas M, Dustdar S, Fahland D, Ferreira DR, Gaaloul W, van Geffen F, Goel S, Günther C, Guzzo A, Harmon P, ter Hofstede A, Hoogland J, Ingvaldsen JE, Kato K, Kuhn R, Kumar A, La Rosa M, Maggi F, Malerba D, Mans RS, Manuel A, McCreech M, Mello P, Mendling J, Montali M, Motahari-Nezhad HR, Zur Muehlen M, Munoz-Gama J, Pontieri L, Ribeiro J, Rozinat A, Seguel Pérez H, Seguel Pérez R, Sepúlveda M, Sinur J, Soffer P, Song M, Sperduti A, Stilo G, Stuel C, Swenson K, Talamo M, Tan W, Turner C, Vanthienen J, Varvaressos G, Verbeek E, Verdonk M, Vigo R, Wang J, Weber B, Weidlich M, Weijters T, Wen L, Westergaard M, Wynn M (2011) Process mining manifesto. In: Daniel F et al (eds) Business process management workshops. Springer, Heidelberg, pp 169–194
- van Eck ML, Lu X, Leemans SJJ, van der Aalst WMP (2015) PM²: a process mining project methodology. In: Zdravkovic J et al (eds) Advanced information systems engineering. Springer, Cham, pp 297–313
- Venkatesh V, Morris MG, Davis GB, Davis FD (2003) User acceptance of information technology: toward a unified view. *MIS Q* 27(3):425–478
- vom Brocke J, Jans M, Mendling J, Reijers HA (2021) A five-level framework for research on process mining. *Bus Inf Syst Eng* 63(5):483–490
- Wagner S, Fernández DM, Felderer M, Vetrò A, Kalinowski M, Wieringa R, Pfahl D, Conte T, Christiansson M-T, Greer D (2019) Status quo in requirements engineering: a theory and a global family of surveys. *ACM Trans Softw Eng Methodol* 28(2):1–48
- Wurm B, Claus R (2024) Reflection on action: the role of process mining for process change. In: Proceedings of the 45th international conference on information systems, Bangkok
- Yin RK (2018) *Case study research and applications: design and methods*, vol 6. Sage, Thousand Oaks
- Zerbato F, Soffer P, Weber B (2021) Initial insights into exploratory process mining practices. In: Polyvyanyy A et al (eds) Business process management forum. Springer, Cham, pp 145–161
- Zerbato F, Soffer P, Weber B (2022) Process mining practices: evidence from interviews. In: Di Ciccio C et al (eds) Business process management. Springer, Cham, pp 268–285
- Zhang P (2013) The affective response model: a theoretical framework of affective concepts and their relationships in the ICT context. *MIS Q* 37(1):247–274
- Zhang X (2017) Knowledge management system use and job performance: a multilevel contingency model. *MIS Q* 41(3):811–840